

Quantum Field Theory I Exercise 8

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1 A fixed point in the ϕ^3 theory

1.1 The linear term of the Wilsonian beta function of g

For any theory in d dimensions, since the kinetic term must have mass dimension d , the dimension of ϕ must be:

$$2[\partial_\mu\phi] = d, [\partial_\mu] = 1 \rightarrow [\phi] = \frac{d-2}{2} \quad (1)$$

This means that the mass dimension of g is:

$$[g] = d - 3[\phi] = d - 3\frac{d-2}{2} = \frac{6-d}{2} \quad (2)$$

Now, since $m = g = 0$ is a fixed point of the theory, the linear terms of the Wilsonian beta for g is actually the Wilsonian gamma γ^g , which is, as we saw, the mass dimension of g :

$$\gamma^g = \frac{6-d}{2} \quad (3)$$

1.2 Perturbation around $d = 6$

We can thus write the Wilsonian beta function as:

$$\beta_{Wil}^g = \gamma_g^g g + \mathcal{O}(g^3) \quad (4)$$

Where we stick to odd powers of g from the antisymmetry of the theory. Setting the dimension to be $d = 6 - \epsilon$, we can write it as:

$$\beta_{Wil}^g = \frac{\epsilon}{2}g + \mathcal{O}(g^3) \quad (5)$$

1.2.1 The fixed point

Perturbing in ϵ and g , we can write this as:

$$\beta_{Wil}^g = \frac{\epsilon}{2}g + \mathcal{O}(g^3) \quad (6)$$

Now, we know that for $d = 6$, meaning $\epsilon = 0$, we get:

$$\beta_{Wil}^g = -\frac{3}{4(4\pi)^3}g^3 + \mathcal{O}(g^5) \quad (7)$$

So, for non-zero ϵ that is still small enough for the approximation to hold, we can write:

$$\beta_{Wil}^g = \frac{\epsilon}{2}g - \frac{3}{4(4\pi)^3}g^3 + \mathcal{O}(g^5) \quad (8)$$

Looking for a non trivial fixed point $\beta_{Wil}^g(g^*) = 0$ we get:

$$g^* = \pm \sqrt{\frac{2(4\pi)^3}{3}\epsilon} \quad (9)$$

Which means that for small enough positive values of ϵ there is a non-trivial fixed point.

1.3 The dimension of ϕ^3 operator

The anomalous dimension of ϕ^3 is then the linear term of the beta function, which can be approximated to the derivative of the beta function at this fixed point:

$$\gamma_{Wil}^g = \left. \frac{d}{dg} \beta_{Wil}^g \right|_{g^*} = \frac{\epsilon}{2} - \frac{9}{4(4\pi)^3}g^{*2} = \frac{\epsilon}{2} - \frac{9}{4(4\pi)^3} \frac{2(4\pi)^3}{3} \epsilon = -\epsilon \quad (10)$$

So, the scaling dimension of ϕ^3 is:

$$\Delta_{\phi^3} = d + \gamma_{Wil}^g = 6 - 2\epsilon \quad (11)$$

Now, comparing this to $d = 6 - \epsilon$, we get that for positive ϵ for which there is a fixed point:

$$\Delta_{\phi^3} = 6 - 2\epsilon < 6 - \epsilon = d \quad (12)$$

So, $\Delta_{\phi^3} < d$. This means that as we go to lower and lower energies, the operator will scale up. This makes it a relevant operator.

Looking around the fixed point, the beta function has a negative derivative:

$$\beta_{Wil}^g \approx -\epsilon(g - g^*) \quad (13)$$

This means that the flow pulls towards the fixed point.

2 The exact renormalization group equation

The exact renormalization group equation is:

$$\begin{aligned} \frac{dg_n(p_1, \dots, p_n)}{d\Lambda} = & \frac{1}{2} \int \frac{d^d p}{(2\pi)^d} \sum_{k, \{n\}} g_{k+1}(p_1, \dots, p_k, p) \times \\ & \times (2\pi)^d \delta^{(d)} \left(\sum_i p_i + p \right) \frac{\partial K^{-1}(p^2)}{\partial \Lambda} g_{n-k+1}(p_{k+1}, \dots, p_n, -p) - \\ & - \frac{1}{2} \int \frac{d^d p}{(2\pi)^d} \frac{\partial K^{-1}(p^2)}{\partial \Lambda} g_{n+2}(p_1, \dots, p_n, p, -p) \end{aligned} \quad (14)$$

For our theory with

$$g_n(\Lambda_0) = \begin{cases} g & n = 3 \\ 0 & \text{else} \end{cases} \quad (15)$$

2.1 Leading order couplings

Starting to generate g_n order by order, in first order of g we have the terms that are generated by the linear term in the above equation. This required

$$g_{n+2} = g_3 \rightarrow n = 1 \quad (16)$$

But since $\langle \phi \rangle = 0$, then $g_1 = 0$ in all orders. Next order is the quadratic term, requiring:

$$g_{k+1} = g_{n-k+1} = 3 \rightarrow n = 4, k = 2 \quad (17)$$

So, in leading order g_4 is generated. Plugging this in gives:

$$\frac{dg_4(p_1, \dots, p_4)}{d\Lambda} = g^2 \frac{\partial K^{-1}((\sum_i p_i)^2)}{\partial \Lambda} \quad (18)$$

This fits the flow of how two 3 point vertices merge to create a single 4 point vertex, where $\sum_i p_i$ is the sum of incoming momenta. Integrating:

$$g_4(p_1, \dots, p_4) = g^2 K^{-1} \left(\left(\sum_i p_i \right)^2 \right) \quad (19)$$

Where we used $g_4(\Lambda_0) = 0$, and the sum $\sum_i p_i$ again matches the sum of incoming momenta. There are three contributing diagrams, for the three channels s, t, u . So, the total g_4 is:

$$g_4(p_1, \dots, p_4) = g^2 [K^{-1}(s) + K^{-1}(t) + K^{-1}(u)] \quad (20)$$

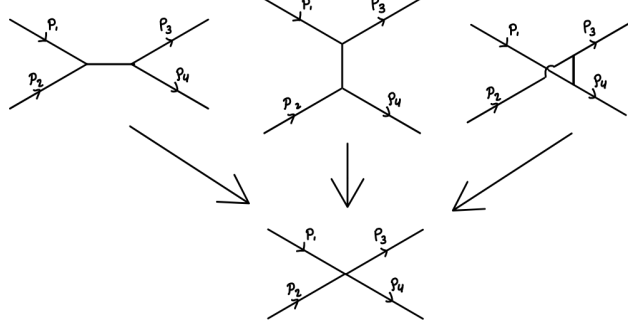


Figure 1: The RG flow of three point vertices to four point vertices. The three channels, s, t, u, all flow to a single four point vertex.

2.2 The sharp cutoff approximation

Using

$$K^{-1}(p^2, \Lambda^2) = \frac{1}{p^2} \Theta(\Lambda^2 - p^2) \quad (21)$$

Since:

$$\frac{\partial K^{-1}(p^2)}{\partial \Lambda} = \frac{d\Lambda^2}{d\Lambda} \frac{\partial K^{-1}(p^2)}{\partial \Lambda^2} = \frac{2\Lambda}{p^2} \delta(\Lambda^2 - p^2) = \frac{\delta(\Lambda - p) + \delta(\Lambda + p)}{p^2} \quad (22)$$

We get:

$$\frac{dg_4(p_1, \dots, p_4)}{d\Lambda} = g^2 \frac{\delta(\Lambda - \sum_i p_i) + \delta(\Lambda + \sum_i p_i)}{(\sum_i p_i)^2} \quad (23)$$

and since $\Lambda > 0$ only one of the delta functions will be non-zero, resulting in:

$$g_4(p_1, \dots, p_4) = g^2 [s^{-1} + t^{-1} + u^{-1}] \quad (24)$$

Now, the other coupling generated by the flow in order g^2 , is the coupling that is generated linearly from g_4 , which is the g_2 coupling:

$$\frac{dg_2(p_1, p_2)}{d\Lambda} = -\frac{1}{2} \int \frac{d^6 p}{(2\pi)^6} \frac{\partial K^{-1}(p^2)}{\partial \Lambda} g_4(p_1, p_2, p, -p) \quad (25)$$

Plugging in the sharp cutoff K^{-1} :

$$\frac{dg_2(p_1, p_2)}{d\Lambda} = -\frac{1}{2} \int \frac{d^6 p}{(2\pi)^6} \frac{\delta(\Lambda - p) + \delta(\Lambda + p)}{p^2} g_4(p_1, p_2, p, -p) \quad (26)$$

Now, from the three channels we get

$$s = (p_1 + p_2)^2 \quad t = (p_1 + p)^2 \quad u = (p_1 - p)^2 \quad (27)$$

Using $d^6 p = \pi^3 p^5 dp$ and then the radial integration variable $p \geq 0$ gives:

$$\frac{dg_2(p_1, p_2)}{d\Lambda} = -\frac{g^2}{2} \int \frac{\pi^3 p^3 dp}{(2\pi)^6} \delta(\Lambda - p) \left[\frac{1}{(p_1 + p_2)^2} + \frac{1}{(p_1 + p)^2} + \frac{1}{(p_1 - p)^2} \right] \quad (28)$$

$$\frac{dg_2(p_1, p_2)}{d\Lambda} = -\frac{\pi^3}{(2\pi)^6} \frac{g^2 \Lambda^3}{2} \left[\frac{1}{(p_1 + p_2)^2} + \frac{1}{(p_1 + \Lambda)^2} + \frac{1}{(p_1 - \Lambda)^2} \right] \quad (29)$$

The g_2 flow generated by g_4 is the flow of a single four point function with a loop.

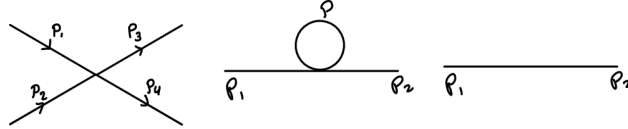


Figure 2: The RG flow of four point vertex to a two point vertex passes through a loop.

2.3 The flow from g^3 to a single vertex

Starting from the triangle loop diagram we saw in tutorial, the flow will start integrating over momentum in a way that will eventually merge two vertices to create a single four point vertex, leaving the third three point vertex alone. Then, the last vertex is also merged to create a five point vertex, before the remaining loop is integrated and the result is back to a 3 point vertex. This can be viewed in figure 3

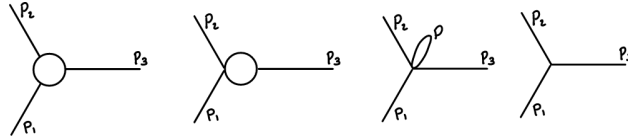


Figure 3: The RG flow from the triangle loop diagram to a single 3 point vertex passing through a four point vertex and a five point vertex.

This can also be viewed from the RG flow. g_3 will generate g_4 from the quadratic term, which in turn will generate g_2 from the linear term and g_5 from the quadratic term. g_5 then again generates g_3 from the linear term.